

How Does Performance Feedback Affect Boundary Spanning in Multinational Corporations? Insights from Technology Scouts

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ABSTRACT As much as prior research has shed light on the boundary-spanning processes of global organizations and their (positive) impact on an MNC's performance, whether, when and how past performance ultimately shapes an MNC's boundary-spanning activities remains an open question in management research. To tackle these questions, we examine the behaviour of technology scouts in global organizations who span organizational and national boundaries to tap into novel knowledge and span internal boundaries to present this knowledge to constituents inside the MNC. Drawing on the behavioural theory of the firm, we propose that the intensity of organizational boundary spanning, the likelihood of spanning national boundaries and the way technology scouts span internal boundaries to engage with members inside the organization are sensitive to how the MNC has performed relative to its aspirations. We support our theoretical expositions with insights gained from interviewing senior industry professionals with direct experience in technology scouting in MNCs.

Keywords: global boundary-spanning process, managing knowledge boundaries, multinational corporations, performance feedback, technology scouts

INTRODUCTION

The ability to tap into knowledge and technologies from outside an MNC's boundary has become a critical organizational activity to compete in environments characterized by rapid technological change (Doz et al., 2001; Gupta and Govindarajan, 2000). The focus of this paper is on the process by which managers in multinational corporations (MNCs) span organizational and national boundaries to tap into external knowledge and span internal boundaries to mobilize such knowledge within the organization (Monteiro, 2015; Monteiro and Birkinshaw, 2017). While this process holds substantial opportunities, it also poses immense challenges. MNCs have to overcome initial

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resistance based on ‘not invented here’ attitudes and prevent their ‘corporate immune system’ from rejecting the knowledge they are trying to bring into the organization (Birkinshaw and Ridderstråle, 1999; Carlile, 2002; Katz and Allen, 1982; Meyer et al., 2011). Moreover, knowledge in rapidly changing environments is characterized by high novelty, which makes managing knowledge across boundaries extremely challenging (Carlile, 2004; Carlile and Rebentisch, 2003; Schotter et al., 2017).

In responding to these challenges, MNCs in a variety of industries have begun to establish dedicated boundary-spanning units, often referred to as technology scouting units, with the mandate to tap into novel external knowledge from other firms around the globe and to present such external opportunities to other parts of the MNC (Birkinshaw and Hood, 2000; Gassmann and Gaso, 2004; Monteiro, 2015). We argue that technology scouts, i.e., the typical title of those working in technology scouting units, are representative of generic boundary spanners in MNCs that are found at a myriad of boundaries between organizational functions, business units, or subsidiaries and the MNC’s headquarters (Carlile, 2002, 2004; Katz and Tushman, 1981; Mudambi and Swift, 2009; Schotter and Beamish, 2011). However, technology scouts are unique in that they are not involved in other day-to-day operations – that is, spanning organizational and national boundaries to seek out external knowledge and spanning internal boundaries to make knowledge available within the organization ‘is the only thing they do’ (Monteiro and Birkinshaw, 2017).^[1] Hence, theorizing about the behaviour of technology scouts can provide unique insights into our broad understanding of boundary spanners in global organizations.

It has long been argued that tapping into knowledge from outside the organization (i.e., spanning organizational boundaries) and from foreign locations (i.e., spanning national boundaries) is associated with significant benefits in the form of innovation and financial performance. Spanning organizational boundaries, for example, allows the addition of distinct new knowledge to an MNC’s knowledge pool and, by adding such global knowledge, an MNC is able to innovate more effectively than its domestic rivals (e.g., Asakawa et al., 2014; Doz et al., 2001; Tushman and Katz, 1980; Tushman and Scanlan, 1981). Prior research has also emphasized the general benefits of spanning boundaries inside MNCs, leading to superior product development (Carlile, 2002, 2004), knowledge management between scientists and managers (Mudambi and Swift, 2009; Tushman, 1977), and initiatives between subsidiaries and their headquarters (Schotter and Beamish, 2011). In a similar vein, prior work has shown that benefits brought by technology scouts stem from both their ability to access new knowledge and their mediating role between the MNC’s external environment and relevant actors (e.g., business managers or R&D units) inside the organization (Monteiro, 2015; Monteiro and Birkinshaw, 2017). Yet, as much as recent research has started to shed light on the boundary-spanning processes performed by technology scouts and their (positive) impact on the MNC’s performance, whether and how the prior performance of an MNC affects boundary spanning activities of technology scouts remains an open question in international management research.

To tackle these questions, we draw upon the behavioural theory of the firm (BTF) and propose that the boundary-spanning behaviours of technology scouts are sensitive to how the MNC has performed relative to its aspirations – that is, the expected levels of

performance the organization seeks to achieve (Cyert and March, 1963). In particular, we suggest that whether a given MNC faces 'harsh times' i.e., when the MNC has performed below its aspirations, or 'slack times' i.e., when the MNC has performed above its aspirations (Greve, 2003b; Levinthal and March, 1981, p. 308) profoundly shapes technology scouts' boundary spanning behaviours.

Central to our theoretical arguments are three key boundary-spanning activities that technology scouts perform and how they are shaped by harsh and slack times: (1) spanning organizational boundaries to engage with the external environment and generate new opportunities (Dollinger, 1984; Santos and Eisenhardt, 2005), (2) spanning national boundaries to reach for 'best in the world' knowledge in foreign locations (Almeida et al., 2002; Cantwell, 1989), and (3) spanning internal boundaries to connect and engage with relevant business and R&D units to mobilize external knowledge within the MNC (Birkinshaw et al., 2017; Carlile, 2002; Meyer et al., 2011; Monteiro and Birkinshaw, 2017).^[2] In particular, we examine how comparisons between aspiration vs. realized performance influence technology scouts' boundary-spanning intensity (i.e., the overall effort technology scouts expend crossing organizational boundaries), and when and how technology scouts' span national boundaries. Further, we analyse how harsh and slack times differently shape the way in which technology scouts span internal boundaries within the MNC to present external solutions to their internal constituents (e.g., business units) and how they go about managing knowledge across internal boundaries in general. We complement our theoretical expositions with examples and insights gained from interviewing senior industry professionals with direct experience in technology scouting in MNCs.

Our paper provides several novel insights into how performance feedback may shape boundary spanning in global organizations. First, we offer a behavioural explanation of technology scouts' venturing beyond firms' boundaries to tap into external knowledge. While previous studies have focused on the knowledge creation potential of boundary spanners (e.g., Carlile, 2002; Doz and Wilson, 2012; Doz et al., 2001; Monteiro and Birkinshaw, 2017), we propose that performance above or below aspirations is an important trigger for boundary spanning in the first place. We suggest that, in both harsh and slack times, technology scouts intensify their efforts in spanning organizational boundaries, but for very different reasons, whereas being close to the aspiration point is likely to lead to inertial behaviour.

Second, we reveal that considering whether a focal MNC is in harsh or slack times provides important behavioural cues as to when technology scouts cross national boundaries to tap into external knowledge and when they combine local activities (in their home countries) with national boundary spanning. We suggest that, in harsh times, to improve performance, technology scouts make costly efforts to span national boundaries but in a directed way and at the expense of other boundary-spanning activities. Conversely, in slack times, boundary-spanning activities will be less directed and more open, which allows technology scouts to combine the spanning of national boundaries with their ongoing activities in their home locations.

We identify a similar pattern when examining how technology scouts span internal boundaries. In harsh times, boundary spanners take into account compensatory trade-offs between external opportunities and internal R&D activities and hence are more

likely to present solutions to their internal constituents as substitutes for internal R&D activities. Conversely, in slack times, resources are readily available, allowing technology scouts to make linkages between internal R&D and external opportunities and to present the complementary nature of external solutions for the organization. Prior studies have emphasized that ambidextrous behaviour – e.g., combining local and foreign knowledge or internal and external knowledge – may lead to higher performance and that the availability of resources is a key moderator for the ambidexterity/performance relationship (Phene et al., 2006; Raisch et al., 2009, p. 395; Rothaermel and Alexandre, 2009). However, our study suggests that performance feedback can be an important antecedent to ambidextrous behaviour in boundary spanning in the first place.

Finally, our study presents a more granular picture about the way technology scouts ultimately manage knowledge across internal boundaries under different conditions. While in harsh times, technology scouts are more likely to emphasize the simple ‘transfer’ or ‘channelling’ of external knowledge towards internal constituents (such as R&D or business units), we propose that in slack times, the emphasis will be more on the ‘translation’ and ‘transformation’ of external knowledge to establish common interests with internal constituents. Prior studies have described those different types of boundary-spanning processes in MNCs and characterized the translation and transformation as value-adding activities by boundary spanners that lead to superior outcomes in product development, knowledge management, and knowledge sourcing (Carlile, 2002, 2004; Monteiro and Birkinshaw, 2017; Mudambi and Swift, 2009). Our study contributes to this stream of research by revealing another contingency – that of being in harsh or slack times – that is likely to affect the way technology scouts manage knowledge across internal boundaries of MNCs.

THEORY

Spanning Organizational, National and Internal Boundaries in MNCs

The idea that organizational members span boundaries to gain access to external knowledge is not new. In fact, the stream of literature on how social systems access and process information from the environment goes back to the open-system perspective in organization theory (March and Simon, 1958; Thompson, 1967). To sustain themselves, organizations have to obtain information from their task environment, but this effort is expensive and time-consuming. Over the past three decades, the boundary-spanning literature in MNCs has diverged into different streams examining distinct levels of analyses. Some studies focused on individuals in R&D departments and how they gather, understand, and transfer external information so that an internal team or unit can use it (Tushman, 1977; Tushman and Katz, 1980), while others examined boundary spanning in teams, focusing on the different ways teams interact with other parts of the organization and with the outside world (Ancona and Caldwell, 1992).

Building on these perspectives, we focus on (1) how managers in multinational corporations cross organizational and national boundaries to tap into external knowledge, and (2) how they span internal boundaries to present the external opportunities they have identified to their internal constituents and, ultimately, bring external knowledge

into the organization (Monteiro and Birkinshaw, 2017). At the heart of this process lies the notion of different types of boundaries (Aldrich and Herker, 1977). Organizational boundaries represent the demarcation between an organization and its environment (Santos and Eisenhardt, 2005), and spanning such boundaries allows access to knowledge that differs from what an organization already knows (Rosenkopf and Nerkar, 2001). While such novel knowledge holds substantial opportunities for organizations, crossing organizational boundaries is far from trivial as it requires overcoming 'not invented here' attitudes from within the organization (Katz and Allen, 1982). National boundaries are important, as seeking external knowledge from foreign locations (in countries outside the MNC's home country, e.g., a Brazilian MNC seeking knowledge from a company in Japan) allows access to 'best in the world knowledge' that often cannot be found in the MNC's home location (Doz et al., 2001; Gittelman, 2007). At the same time, spanning national boundaries is costly since knowledge tends to be localized, embedded in practice, and invested in by its originators (Almeida et al., 2002; Carlile, 2002; Meyer et al., 2011).

A final set of boundaries lies within the organization. Once identified, external (and potentially foreign) knowledge needs to be integrated and used by organizational members (e.g., by a R&D unit in product development or a business unit in sales), which requires 'selling' the knowledge to internal business or R&D units (Monteiro and Birkinshaw, 2017). This can be challenging as parts of the MNC may resist novel external knowledge because it does not fit their prevailing knowledge pool or may reject such knowledge owing to fears of cannibalizing existing competencies or revenue streams (Birkinshaw and Ridderstråle, 1999; Mudambi and Swift, 2014).

The opportunities and challenges stemming from boundary spanning are particularly relevant in knowledge-intensive environments, where technological opportunities emerge rapidly; existing competencies quickly become obsolete; and intense rivalry and competition among industry participants prevail (D'Aveni et al., 2010). In such environments, for example, telecommunications and pharmaceuticals, spanning organizational and national boundaries to obtain novel knowledge is considered pivotal to an organization's competitive advantage (Banker et al., 2013; Bierly et al., 2009; Chesbrough et al., 2006). At the same time, managing knowledge across boundaries in those environments is extremely challenging as the underlying knowledge is characterized by novelty (e.g., rapid technological advancements or changing customer requirements), strong differences between the MNC and the potential source of knowledge (e.g., due to specialization or size effects) and dependencies (i.e., the MNC relies to some extent on external actors to ultimately make productive use of the external knowledge) (Carlile, 2004; Carlile and Reberntsch, 2003). Solving all these challenges is of utmost importance for today's MNCs (Schotter et al., 2017).

Technology Scouts in Global Organizations

A possible solution is specialization by assigning specific individuals or teams as boundary spanners who 'gather and understand external information and subsequently translate this information into terms that are meaningful and useful to local colleagues' inside the organization (Tushman and Katz, 1980, p. 1073). In recent decades, some MNCs

have begun to more systematically organize their boundary-spanning activities by creating foreign subsidiaries, or dedicated units, often referred to as scouting units, with a specific boundary-spanning mandate (Birkinshaw, 1996; Birkinshaw and Hood, 2000; Gassmann and Gaso, 2004). Such boundary spanners, henceforth technology scouts, are now found in many industries, including telecommunications (e.g., British Telecom), pharmaceuticals (e.g., Merck and Co., Bayer), and electronics (e.g., Siemens).

It is important to note that technology scouts span organizational and national boundaries to tap into external knowledge as well as internal boundaries to present such opportunities to other parts of the MNC (e.g., Doz et al., 2001; Patel and Vega, 1999; Von Zedtwitz and Gassmann, 2002). They are also more than just brokers, i.e., actors in organizations positioned between structurally separated and unconnected entities (Burt, 1992). Burt sees brokers operating with a ‘*tertius gaudens*’ orientation, in which a broker can create value through superior access to information and greater opportunities to exercise control. Conversely, we concur with Monteiro and Birkinshaw (2017) and consider technology scouts to have a ‘*tertius iungens*’ orientation, ‘a strategic, behavioral orientation toward connecting people in one’s social network by either introducing disconnected individuals or facilitating new coordination between connected individuals’ (Obstfeld, 2005, p. 102). Hence, they are boundary spanners as they go well beyond their structural position and connect, introduce and coordinate previously unconnected parties for mutual benefits (Fleming and Waguespack, 2007).^[3]

We believe technology scouts are an important representation of generic boundary spanners in MNCs that are known to be situated at the intersection of different types of boundaries, including R&D teams and their external environment (Katz and Allen, 1982), scientists and business managers (Mudambi and Swift, 2009), engineering and manufacturing (Carlile, 2002, 2004), or subsidiaries and the headquarters (Schotter and Beamish, 2011). Indeed, Monteiro and Birkinshaw (2017) have characterized technology scouts as an ‘extreme example’ of boundary spanners mediating between the organization and its external environment as well as spanning boundaries within the organization. This is because, unlike other boundary spanners in the organization, technology scouts do not have other day-to-day operating roles they need to execute and boundary spanning is what they do for the organization.^[4] By analogy, technology scouts could be to boundary spanning what Saxenian’s (2007) ‘Global Argonauts’ represent to international entrepreneurship – a very ‘pure’ and visible form of an important but often difficult to observe phenomenon. Theorizing about the behaviour of technology scouts therefore broadens our understanding of boundary spanning in global organizations.

Boundary Spanning and Performance

Studies have shown that MNCs that span boundaries to access external knowledge innovate more effectively than their domestic rivals and in general have advantages over industry peers (e.g., Asakawa et al., 2014; Dollinger, 1984; Doz et al., 2001; Monteiro, 2015). This is because adding novel pockets of knowledge to an organization is directly associated with a firm’s innovative and financial performance (e.g., Hagedoorn and

Schakenraad, 1994; Kale et al., 2002; Laursen and Salter, 2006; Leiponen and Helfat, 2009).

In a similar vein, research focusing on the role of boundary spanners has shown that spanning external boundaries is related to innovation and performance benefits (Tushman and Katz, 1980; Tushman, 1977). Studying more systematically the boundary-spanning processes undertaken by technology scouts has also revealed distinct benefits of spanning internal boundaries to connect external knowledge to R&D or business units to make external knowledge available inside the MNC (Monteiro and Birkinshaw, 2017). This follows the idea that boundary spanners can better manage tensions between separated entities or groups, hence leading to superior outcomes and improved performance (Carlile, 2002, 2004; Mudambi and Swift, 2009; Schotter and Beamish, 2011; Tippman et al., 2017).

Yet, while research has made notable progress in linking boundary spanning to innovative and financial performance outcomes, we know much less about the recursive relationship, that is, whether and how the prior performance of an organization ultimately shapes the boundary-spanning behaviours of technology scouts.

A Performance Feedback Perspective on Boundary Spanning by Technology Scouts

A well-established stream of literature built on the behavioural theory of the firm (BTF) examines how performance feedback affects an organization's strategic decision-making (Cyert and March, 1963; March and Simon, 1958). We extend this literature by suggesting that performance feedback may significantly affect boundary-spanning activities by technology scouts.

The BTF characterizes firms as goal-directed systems that use simple operating procedures and heuristics to adapt their behaviour. Organizational goals are important, as they are considered relevant by resource-allocating managers and lead to the formation of aspirations, i.e., expected levels of performance that organizations aspire to achieve.^[5] Managers compare their firm's aspirations with actual performance outcomes, leading to favourable (performance outcomes above aspirations) or unfavourable (performance outcomes below aspirations) comparisons (Cyert and March, 1963; Levinthal and March, 1981).^[6]

In the majority of performance feedback studies, the crux of the argument is that by comparing aspirations with actual performance, managers find themselves in situations of either harsh times (performance outcomes below the aspiration point) or slack times (performance outcomes above the aspiration point) (Levinthal and March, 1981, p. 308) and being in either harsh or slack times shapes the decision-making of resource-allocating managers (Greve, 2003b). Supporting this idea, prior research has shown that the aspiration point functions as a 'master switch' that influences a variety of strategic behaviours and choices related to innovation, including organizational risk taking (Bromiley, 1991), R&D spending (Chen and Miller, 2007), capital investments, new product development (Greve, 2003a), acquisitions (Iyer and Miller, 2008), and organizational change (Greve, 1998). Interestingly, however, studies have given much less attention to examining how performance-aspiration comparisons are likely to affect the activities of

boundary spanners such as technology scouts. In other words, most studies have focused on the performance outcomes of spanning boundaries and performance as a result of boundary spanners' actions (e.g., Asakawa et al., 2014; Doz et al., 2001; Monteiro, 2015). However, we believe that studying the inverse relationship – how performance may act as an input to technology scouts' behaviour – can generate new insights into how the boundary-spanning process unfolds in global organizations.

We focus on how performance feedback shapes three critical activities by technology scouts in MNCs.^[7] First, managers may face important decisions on how much emphasis they should place on crossing organizational boundaries in the first place, and organizational boundary-spanning intensity characterizes the overall efforts by technology scouts to seek out external knowledge (Dollinger, 1984). Second, technology scouts can engage with local actors or span national boundaries to tap into 'best in the world' knowledge in foreign locations (Doz et al., 2001; Gittelman, 2007). Finally, we examine how technology scouts manage knowledge across internal boundaries and engage with internal business or R&D units to mobilize external knowledge for the organization (Meyer et al., 2011; Monteiro and Birkinshaw, 2017).

For each activity, we provide examples and illustrations which we derived by interviewing senior industry professionals from different industries (such as telecommunications, medical devices and pharmaceuticals) and with direct experiences in global boundary spanning and technology scouting. Specifically, between 2011–15, we conducted 15 in-depth interviews with managers in European and North American MNCs (including Bayer, Johnson & Johnson, Merck & Co, and Novo Nordisk) on the MNCs' sites, at technology conferences such as the Wharton Mack Institute's Technology Conference or via phone. The majority of the interviewees had titles directly associated with technology scouting or were the managers of a technology scouting (or business development) unit. Some interviewees were senior managers of internal R&D and business units, which represented the 'customers' of technology scouts' boundary-spanning efforts. Finally, we also interviewed top management executives responsible for the overall externally oriented activities in their respective multinational organizations to obtain a more strategic picture of the roles technology scouts play in global organizations.^[8]

Spanning Organizational Boundaries in Harsh and Slack Times

An important element of boundary spanning by technology scouts is its intensity, that is, the overall effort exerted by boundary spanners to cross boundaries (Dollinger, 1984). Here we focus on the intensity with which technology scouts cross organizational boundaries that potentially hold substantial opportunities due to the access to new knowledge, but that is not trivial, as 'not invented here' attitudes may result in the entrenchment of organizations (Katz and Allen, 1982; Rosenkopf and Nerkar, 2001; Teece, 1996). Hence, when spanning organizational boundaries, technology scouts' first challenge may arise from persistent negative attitudes toward ideas external to their organizations, which may prevent MNCs from becoming more open and from seeking out external ideas in the first place. An executive responsible for worldwide scouting and licensing at a large pharmaceutical firm described the challenge as follows: *'It is clearly important to identify high value external opportunities that will allow us to grow. [...] But the question*

we face is how much should we emphasize our externally oriented activities. . . ? A major problem we have is that these activities are not yet on a level playing field with our other, more, traditional ways of doing research and development. . . This is an ongoing problem that has to be worked through . . . We are not sufficiently open to external ideas'.

Drawing on the BTF, we suggest that the intensity (i.e., the overall effort made) with which technology scouts span organizational boundaries is sensitive to whether the MNC is facing harsh or slack times.

In harsh times. Models of performance feedback predict that an organization is more likely to take action when it fails to reach its aspirations (e.g., Greve, 2003b; Levinthal and March, 1981). In harsh times, managers face short-term pressures to improve their performance because they are held accountable for prior decision making (Cyert and March, 1963). Unable to conceal below-aspiration performance from internal and external stakeholders, managers are pressed to find a solution. However, such solutions may not be readily available within an organization. We argue that spanning organizational boundaries may allow MNCs to access 'ready-to-use' solutions that, in some cases, would require substantial time to develop internally (Hagedoorn, 1993). This time investment is particularly relevant in knowledge-intensive environments, where solutions to solve below-aspiration performance tend to be needed immediately. Hence, in harsh times, technology scouts are likely to intensify the spanning of organizational boundaries to connect to solutions that remedy the organization's problems. This approach is consistent with findings of research in international business showing that knowledge flows in MNCs can be explained by demand-driven, problem-motivated processes (Monteiro et al., 2008).

A senior manager responsible for technology scouting at Merck provided an illustration of this phenomenon. He recalled that in 2004, the withdrawal of a pain-killer drug led to substantial legal provisions, which had dramatically decreased financial performance (an absolute decrease of \$700 million in net income, about a 12 per cent decline compared to 2003).^[9] The manager emphasized how being in harsh times exacerbated the need to span organizational boundaries: *'The company was at a real tipping point. . . To compensate for the decline, we had to get better at leveraging the external environment. . . To me, the alternatives were stark. Either we were successful or Merck would have to downsize. The changes that we needed were to prop up 30–40 per cent more of the entire pipeline with technologies [sourced from] outside Merck. (Example 1)*

In the following years, technology scouts at Merck crossed organizational boundaries more aggressively. One technology scout remembered that *'Merck during that year [2004] faced a situation that was not business as usual. . . There was an overall understanding that we needed a quick fix'.* He recalled that, at the time, the technology scouting organization expanded substantially to *'connect to and acquire external research. . . We were told to participate in conferences and get out [of Merck's R&D facilities] to identify promising projects for short-term revenue generation. . .'* (Example 2)

As that example also illustrates, harsh times require the clear articulation and codification of an underlying problem (Zollo and Winter, 2002). In Merck's case, codification was important to establishing new routines and standard operation procedures in the firm's boundary-spanning activities. Hence, when performance is below aspirations, externally oriented activities are more likely to be formalized, for example, by

establishing a dedicated technology scouting unit. Indeed, prior work has suggested that boundary-spanning roles become formalized when managers realize and codify crucial environmental contingencies (Aldrich and Herker, 1977), as is more likely to happen in harsh times. Overall, we posit that the wider the gap between an MNC's performance and its aspirations, the more likely it is that MNC will incept a technology scouting unit with the mandate of spanning organizational boundaries and the more likely technology scouts intensify spanning organizational boundaries.^[10] We propose:

Proposition 1a: The further an MNC's performance is below its aspirations, the more intense technology scouts' efforts will be to span organizational boundaries.

Proposition 1b: The further an MNC's performance is below its aspirations, the more likely the MNC will formalize or establish a dedicated boundary-spanning unit.

In slack times. At first glance, performing above aspirations might seem to have no direct effect on whether technology scouts span organizational boundaries because they do not face immediate performance pressures and may be content with the status quo. However, most MNCs operate in knowledge-intensive industries, i.e., environments in which competitive advantages based on old resource positions are continuously challenged and advantages, even of leading high-performing firms, are considered temporary (D'Aveni et al., 2010; Winter, 2003).

To maintain their position in such knowledge-intensive environments, managers need to continuously generate innovative moves, even if they have persistently out-performed their aspirations (Kirzner, 1973). Given that, in such environments, innovations tend to emerge beyond firm boundaries (Chesbrough et al., 2006), an important way to keep ahead is to systematically seek new knowledge through technology scouts. Spanning organizational boundaries through technology scouts, hence, can become a moving target, as all firms continue to make innovation-related investments to avoid falling behind in the future – an outcome that has been described as the 'Red Queen' effect (Banker et al., 2013). We expect that such behaviour will more likely transpire in slack times for the following reasons.

First, technology scouts in slack times do not have immediate problems to solve and are under less scrutiny by external stakeholders to find urgent solutions. This may allow them to focus on 'next problems', allowing technology scouts to adopt a more opportunistic or proactive behaviour (Cyert and March, 1963; Monteiro and Birkinshaw, 2017). This explanation is consistent with the idea that managers are more likely to promote entrepreneurial opportunities in their organizations if they are satisfied with their current performance situation (De Clercq et al., 2011; Simsek et al., 2007). Hence, in slack times, technology scouts are most likely more able to span organizational boundaries to tap into external knowledge.

As an example, a senior executive manager responsible for external business development for Johnson & Johnson emphasized the need to opportunistically span organizational boundaries in slack times: '*No matter what we do within our in-house R&D, the rest of the world is always doing more*'. He also cited the need '*to keep connecting with those start-ups even if we are doing very well. . . In 2011, the medical device unit had clearly outperformed the market over a*

longer time span. . . Still, even when we were doing well . . . there was an increasing appreciation that 99.9 per cent of the science is beyond our labs. . . During this time, we were asked to examine what was available [externally] in an exhaustive way. There was a general fear to not miss anything really important' (Example 3a).

Second, such proactive or opportunistic boundary spanning by technology scouts may be catalysed by the presence of slack, that is, excess liquid resources or operational resources beyond those required in the short term (Singh, 1986). In general, 'success tends to breed slack' (Cyert and March, 1963, p. 189), suggesting that slack tends to be present when an organization performs above aspirations (Pitelis, 2007). Excess resources are fungible (Levinthal and Wu, 2010) and thus can easily be channelled toward boundary spanners.^[11] A technology scout responsible for sourcing external technologies for a global pharmaceutical leader in the diabetes market highlighted how the presence of slack helped to intensify spanning organizational boundaries. '*We are very happy with the recent diabetes products we put on the market and we remain very, very successful. . . We are now in a mode where things are less tight and we engage in larger efforts to learn what is out there. . . I love that this unit [diabetes] never rests, which makes my work so exciting. Ultimately, it allows me to generate much more opportunities'* (Example 3b).

Finally, outsiders may perceive organizations in slack times as being more successful, which makes them more attractive to potential partners to begin with (Ahuja, 2000). Being exposed to a large variety of opportunities may guide firms to more systematically organize technology scouts and formalize their activities (e.g., in the form of a dedicated unit). For example, an executive manager of Johnson & Johnson's corporate development program noted: '*We realized that we needed a more systematic way to deal with external opportunities but also opportunities approaching us and we began to substantially increase budgets for such initiatives'* (Example 4).

Since MNCs in slack times will be motivated to maintain their competitive advantage and provide boundary spanners with adequate resources to seek out external knowledge, global boundary-spanning activities can intensify when performance exceeds aspirations. In a similar vein, Pitelis (2007) argued that in environments where inter-firm competition is high, slack resources may be used to stay ahead of rivals. We thus propose:

Proposition 1c: The further an MNC's performance is above its aspirations, the more intense technology scouts' efforts will be to span organizational boundaries.

Proposition 1d: The further an MNC's performance is above its aspirations, the more likely the MNC will formalize or establish a dedicated boundary-spanning unit.

Close to the aspiration point. Taken together, the above propositions reveal two interesting facets of how performance feedback affects the intensity with which technology scouts are likely to span organizational boundaries. First, the further MNCs are above or below aspiration levels, the more intense the technology scouts' organizational boundary-spanning efforts, but for very different reasons and with qualitatively different triggers (to solve a problem in harsh times, to find new opportunities in slack times). Second, in MNCs that are close to their aspiration levels, technology scouts are more likely remain inert and less likely to intensify their organizational boundary-spanning efforts.

In such MNCs, technology scouts have neither the pressure (as those in harsh times) nor the slack resources (as those in slack times) to intensify those activities. More formally, we propose:

Proposition 1e: When an MNC performs close to its aspiration point, technology scouts less likely intensify efforts to span organizational boundaries.

Proposition 1f: An MNC performing close to its aspiration level is less likely to formalize or establish a dedicated boundary-spanning unit.

Spanning National Boundaries in Harsh and Slack Times

While, in the previous section, we focused on the intensity of the efforts made by boundary spanners in crossing organizational boundaries, here we focus on another type of boundary spanning: how harsh and slack times are likely to affect whether and how technology scouts span national boundaries to connect and engage with foreign firms.

An important challenge for boundary spanners lies in the preference of organizations for local knowledge (Belderbos et al., 2013; Nobel and Birkinshaw, 1998).^[12] Tapping into such knowledge can be easier and more cost-effective because being close to the knowledge source facilitates understanding and integrating the knowledge, particularly when such knowledge is tacit and embedded in a local context (Galunic and Rodan, 1998; Tallman and Chacar, 2011). Conversely, spanning national boundaries (i.e., tapping into knowledge from firms in different countries) entails substantial costs as it requires crossing into unfamiliar local contexts (Castellani et al., 2013; Kranenburg et al., 2014). A technology scout in a US-based pharmaceutical firm explained the challenges of spanning national boundaries as follows: *‘I think it is fair to say we still have a very American-centric view of the world and that makes it harder for me. . . . Sometimes I am not sure if there is a real interest and commitment towards these regions [Australia, Asia]. . . . One reason is that 70 per cent of biotech industry is in the US and that’s the way it is. . . . Things are not as concentrated in Asia and Australia and this requires more “digging” from my side. . . . Often, the approaches in my regions are the ones that have not been explored by us anyway’.*

The quote reveals that, while challenging, spanning national boundaries may hold substantial value as it allows organizations to access novel elements of knowledge and ‘best in the world’ knowledge, which may not be available in the MNC’s home country (Doz et al., 2001; Gittelman, 2007). We next examine how being in harsh or slack times affect if and how technology scouts span national boundaries.^[13]

In harsh times. Understanding that, in harsh times, MNCs are motivated by existing (and short-term) problems suggests that the starting point for boundary spanning by technology scouts will typically be in the MNC’s home country, where it is embedded and into which it can tap quickly with little extra cost (Belderbos et al., 2013; Cyert and March, 1963). A pharmaceutical technology scout highlighted that, in harsh times, he is unlikely to cross national boundaries: *‘solutions sought [in harsh times] often tend to be not far from the locations where we were active before. This is easier for me as I can collect the information rapidly and just deliver what they want (Example 5).*

However, as performance falls further below aspirations, boundary spanners may be pressured to consider knowledge that is only available in foreign locations (Gittelman, 2007; Hannigan et al., 2015), which requires spanning national boundaries. It is well-established that, in harsh situations, decision makers may seek solutions from which they expect to disproportionately benefit, leading them to make riskier choices (Das and Teng, 2001). Hence, when performance falls further below aspirations, we expect technology scouts to span national boundaries to tap into novel and 'best in the world' foreign knowledge, which is risky but that can also have a substantial upside. As an example, following near-bankruptcy in the late 1970s, the Big 3 automakers reached out explicitly to Germany, far from their home location as Germany was considered one of the global hotbeds in the automotive industry at the time (Hannigan et al., 2015).

Two caveats apply to the idea that technology scouts are more likely to span national boundaries if the performance-aspiration gap has become too wide. First, in harsh times, resources to support technology scouts are likely scarcer. During those periods, managers expend considerable effort attending to pressures from internal and external stakeholders, which constrains resources (Abrahamson and Park, 1994). At the same time, substantial efforts by technology scouts are needed to span national boundaries and, in most cases, boundary spanners need to be present (e.g., having face-to-face contact) to comprehend opportunities emerging in foreign locations (Storper and Venables, 2004). Knowledge in foreign countries is also characterized by particularly strong differences between the MNC and the potential foreign source of knowledge (due to strong specialization and localization) and dependence (i.e., the foreign knowledge source is embedded in a local context and the MNC needs to rely extensively on the knowledge source to transfer and use that knowledge within the MNC), exacerbating the costs of managing knowledge across national boundaries for technology scouts (Carlile and Rebentisch, 2003; Inkpen and Beamish, 1997). It is hence likely that technology scouts need to make compensatory trade-offs in their boundary-spanning efforts and balance staying inside national boundaries or spanning national boundaries. The result is that, in harsh times, spanning national boundaries may come at the expense of other activities by technology scouts, which is the reason we expect national boundary spanning to be punctuated and, in many cases, occur only as a one-off activity.

Second, even if technology scouts span national boundaries, such activities are likely to remain directed and constrained by the organization's existing problems. This limitation suggests that, in harsh times, boundary spanners most likely seek out predefined 'needs' and have a clear understanding of where to look for relevant knowledge to remedy the performance-aspiration gap (Monteiro, 2015). As a result, the scope of spanning national boundaries will be limited since the range of possible locations and solutions is predefined. A technology scout from a German MNC emphasized the directed nature of distant boundary-spanning activities in harsh times: *'The only good thing in those circumstances [harsh times] is that I get very clear instructions from my business heads where to look [in China]. . . . They [the business unit contacts] will quickly tell me what they want and not want* (Example 6).

Overall, we expect that performance falling far below aspirations can lead to punctuated national boundary spanning by technology scouts. We propose:

Proposition 2a: The further MNCs perform below aspirations, the more likely it is that technology scouts will span national boundaries in a punctuated way.

Proposition 2b: In harsh times, technology scouts have a predefined and narrow mandate to span national boundaries.

In slack times. Positive performance feedback induces repeated behaviour (Cyert and March, 1963), which would suggest that, in slack times, boundary spanners are likely to remain confined within their national borders. However, similar to our argument about organizational boundary spanning in slack times, there is reason to believe that slack times may also allow technology scouts to span national boundaries and tap into knowledge in foreign countries.

We identified that managers in slack times do not have immediate problems to solve and can focus on ‘next problems’, which encourages experimentation (Nohria and Gulati, 1996; Sidhu et al., 2004). Hence, technology scouts in slack times are free from the constraints of solving urgent operational problems and can experiment with spanning national boundaries by going to countries in which the MNC has no operating activities (Doz et al., 2001). In a similar vein, researchers have suggested that decision makers with longer planning horizons (i.e., absent urgent problems) may engage in risk-seeking behaviour, indicating that technology scouts in slack times may span more ‘risky’ (i.e., national) boundaries (Das and Teng, 2001).

Spanning national boundaries is also facilitated by the presence of excess resources in slack times. These resources are particularly important given the costs of being physically present in a foreign country and the underlying properties of knowledge (novelty, difference and dependence), which make the management of knowledge at such boundaries extremely challenging (Carlile, 2004; Carlile and Rebentisch, 2003). It stands to reason that, with the availability of excess resources to support such costly activities in slack times, technology scouts are more likely to experiment and span national boundaries.

Supporting these ideas, a global R&D manager (similar to a technology scout) at a German MNC explained that, during slack times, spanning national boundaries was easier: *‘My business unit has really outperformed the last years, so we have greater freedom to explore different geographies. . . . As an example, they allowed me to pursue quite a few exploratory programs with university professors all over the world, including China, as they are convinced that there are many yet to be discovered targets and, in general, therapeutic improvements. . . . [I]n the last few years, the unit was particularly willing to tap into the “unknown” with me (Example 7).*

In contrast to harsh times, however, we do not expect national boundary spanning to be punctuated and to come at the expense of technology scouts’ activities within their home locations. Instead, in slack times, boundary spanners can more likely combine their activities within and across national boundaries to tap into external knowledge. Moreover, the lack of an underlying problem suggests that, in slack times, technology scouts are much more open to spanning boundaries all around the globe. This view expands on the ambidexterity perspective, which highlights that organizations enjoy performance benefits when combining local and foreign knowledge and that slack resources are an important moderator for the ambidexterity-performance relationship

(Raisch et al., 2009). Our arguments reveal that the reverse relationship may also hold; slack times more likely foster ambidextrous behaviour (by boundary spanners) in the first place. We propose:

Proposition 2d: The further MNCs perform above aspirations, the more likely it is that technology scouts will span national boundaries.

Proposition 2e: In slack times, technology scouts have broad and open mandates to span national boundaries.

Spanning Internal Boundaries in Harsh and Slack Times

So far, we have focused on the externally oriented activities of boundary spanners. However, connecting to external organizations, possibly in foreign locations, is just one part of the boundary-spanning process undertaken by technology scouts. The other equally important part is how they manage internal boundaries by presenting the opportunities they have externally identified to internal R&D units, business units or the top management and how this process is shaped by whether the MNC is in harsh or slack times.

Boundary spanners must not only seek out external knowledge but also sell and integrate it into the organization (Monteiro, 2015). This persuasive effort can be challenging since internal R&D competes with boundary spanners for organizational resources. and often MNCs resist the external solutions that boundary spanners present (Birkinshaw and Ridderstråle, 1999; Raisch et al., 2009). Hence, a major task for technology scouts lies in spanning internal boundaries and convincing organizational members and groups (such as a business or R&D unit) to apply and integrate external knowledge (Birkinshaw et al., 2017; Meyer et al., 2011; Monteiro and Birkinshaw, 2017; Mudambi, 2011). One senior executive responsible for the technology scouting activities in an MNC explained this challenge: *‘One of the challenges we face is that our scouts can travel around the world and spend endless time digging into new external opportunities, but if franchises [the business units] are then not aligned with this approach, it may be an inefficient process overall. . . In some cases, they [the business units] still give preferences to internal projects over the external opportunities we present. . . This can be frustrating because you can get people inside our R&D organization who are very dogmatic’.*

Research has emphasized that, in general, external knowledge can be a substitute for or complement to internal R&D activities (Pisano, 1990; Veugelers, 1997). Hence, when spanning internal boundaries to internal constituents, technology scouts may either emphasize external knowledge as a substitute for or complement to internal R&D efforts. However, in exploring this tension, previous research has not yet probed whether MNCs are in harsh or slack times. Next, we discuss how harsh and slack times ultimately shape how technology scouts manage knowledge across internal boundaries.

In harsh times. Global boundary spanning to tap into new knowledge is generally a costly process requiring significant financial, technological, organizational, and managerial resources (Castellani et al., 2013; Levinthal and March, 1981). For boundary spanners, this process will entail identifying possible technological solutions around the globe, translating and disseminating the search results within the MNC, and ultimately internalizing the externally sourced technologies (Monteiro, 2015).

Given the previously outlined resource constraints in harsh times and that R&D budgets tend to remain stable (e.g., as a fixed percentage of revenue), managers would need to make compensatory trade-offs in allocating resources between R&D and the activities by technology scouts. We therefore expect that, in harsh times, technology scouts emphasize external knowledge as a possible alternative and substitute for internal R&D when they span internal boundaries to present identified knowledge to R&D units, business units or managers.

In harsh times, organizations are also characterized by conflict because they have failed to achieve the performance to which they aspired, leading to tension among the MNC's coalitions vying for scarce resources (Cyert and March, 1963). While, in principle, organizations try to avoid conflict (Pitelis, 2007), once it is present, boundary spanners may deliberately depart from existing internal ways of solving problems and seek alternative solutions. In such cases, technology scouts may intentionally utilize this state of tension and present external solutions as substitute for internal initiatives. A director of licensing and acquisitions and technology scout at a pharmaceutical MNC highlighted how, during harsh times, her boundary-spanning efforts emphasized how external solutions can be substitutes for internal R&D activities. *'Based on [...] performance outcomes, we cut programs on a daily basis and they are very sensitive to changes in their KPIs (key performance indicators). It is also true that some units, as a result of underperformance, had substantial lay-offs, so they are in dire need of external opportunities, which meant more work for me. In many cases, we measured the FTE [full time employee] rate and compared it against our internal one. As you can imagine, this is not a great conversation [with the internal R&D units] as it usually ends up calculating scientists' costs from our internal R&D operations versus potential external partners (Example 8).'*

Overall, we suggest that, in MNCs performing below aspirations, technology scouts spanning internal boundaries will present external solutions as a substitute for internal R&D. More formally, we propose:

Proposition 3a: In harsh times, when spanning internal boundaries, technology scouts are more likely to present external knowledge as a substitute for internal R&D activities.

The above arguments indicate that harsh times may also influence the way in which technology scouts engage with other units inside their organizations. Research has distinguished various ways by which boundary spanners mediate between the external environment and the internal organization, which range from mere communication, channeling, and transfer of knowledge to translating and transforming external knowledge for internal constituents (Carlile, 2002, 2004; Monteiro and Birkinshaw, 2017). Transferring or channeling knowledge is grounded in information processing and communication of facts. Conversely, translating and transforming knowledge requires technology scouts to substantially engage with their internal constituents and create common interests to have external knowledge mobilized internally (Carlile, 2004; Monteiro and Birkinshaw, 2017).

We identified that, in harsh times, technology scouts are more likely to present external solutions as substitutes for internal initiatives, so one would expect that boundary spanners would focus on transferring 'ready-to-use' external knowledge into the

organization but without engaging in more value-added (but costly and time-consuming) boundary-spanning processes such as translating or transforming. As the technology scout revealed in Example 8, in harsh times, she focused on transferring external information and communicating facts to the organization's internal R&D constituents. Indeed, such behaviour fits the transferring and channeling type of boundary spanning (Carlile, 2004; Monteiro and Birkinshaw, 2017). We suggest that, when taking into account prior performance, harsh times (i.e., performance below aspirations) will lead technology scouts to span internal boundaries predominantly through the channeling and transferring of external knowledge towards their internal constituents. More formally, we propose:

Proposition 3b: In harsh times, when spanning internal boundaries, technology scouts are more likely to emphasize the transfer and channelling of external knowledge into the MNC.

In slack times. In contrast to harsh times, in slack times, technology scouts are less likely to be driven by substitution pressures. In slack times, MNCs enjoy excess managerial and financial resources that they have accumulated but have not yet consumed or incorporated into organizational processes (Cyert and March, 1963). If managers follow standard operating procedures in allocating resources to R&D (e.g., as a percentage of revenue), performance above aspirations may allow firms to allocate some resources to global boundary-spanning initiatives while still maintaining their level of investment in internal R&D. Hence, technology scouts in slack times are under less pressure to make trade-offs between internal R&D and external knowledge and have more time to sell external innovative ideas within the organization (Mudambi and Swift, 2009). Moreover, in slack times, boundary spanners will find it easier to draw on the attention of managers who have no immediate problem to solve. Having more managerial attention available increases the probability that boundary spanners can champion external activities as a complement to internal R&D within the organization, and look at global external technologies in a more complementary way (Crossan and Apaydin, 2010; Dutton et al., 2001).

In addition, in slack times, there is likely less conflict between coalitions within the MNC (Cyert and March, 1963). Given that organizations tend to eschew conflict, boundary spanners will avoid presenting external opportunities as a substitute for internal R&D initiatives to their internal constituents. Instead, it becomes more likely that the technology scouts make a conscious effort to underscore the complementarity nature between what they have found and internal R&D initiatives. This reduces tensions which could emerge from novel external knowledge being brought into the organization (Mudambi and Swift, 2009). A manager responsible for external R&D units (comparable to technology scouts) at a German MNC revealed this behaviour during slack times. *'I still prefer the mode when things are going well as we have greater freedom to learn what is out there. The good thing is that they are interested in how what I find ultimately will help their [internal] R&D initiatives and help the [business] unit grow. . . I showcase to them [the R&D unit managers] how what I find actually helps them achieve their goals (Example 9).'*

The example demonstrates that, in slack times, technology scouts may emphasize the potential complementary benefits of their activities for the internal R&D units. This once again reflects the ambidexterity perspective, which suggests that organizations enjoy performance benefits when combining both internal and external R&D (Rothaermel and Alexandre, 2009). Our arguments posit that the reverse relationship may also hold; slack times are more likely to foster ambidextrous behaviour by allowing technology scouts to explore complementarities of internal R&D and external opportunities. We propose:

Proposition 3c: In slack times, when spanning internal boundaries, technology scouts are more likely to present external knowledge as a complement to internal R&D activities.

The availability of slack also increases the likelihood that technology scouts can emphasize more value-added boundary-spanning activities, such as the translation and transformation of external knowledge to more effectively integrate it into the organization (Monteiro and Birkinshaw, 2017). The translation of knowledge requires some manipulation by scouts so that it can be better understood inside the MNC (Carlile, 2004; Tushman and Scanlan, 1981). In particular, transforming knowledge requires technology scouts to recognize novel dependencies between external knowledge and internal R&D activities and define common interests with the internal units (Carlile, 2002; Monteiro and Birkinshaw, 2017). Both translation and transformation by technology scouts are costly, and hence we expect such behaviour to more likely transpire in slack times. A technology scout at Merck discussed his interactions with one of the successful business units and revealed that working with that unit required substantial listening and transforming knowledge for the internal R&D organization: ‘... one of the most critical things with them is making sure that there is concurrence between what I am looking for and what they [the business unit] need. That’s where it starts. Interestingly, the successful franchises are the ones more open to such conversations. In such cases, I try to discuss the usefulness of the external technologies with the unit and try hard to reconcile what I found with the franchise’s areas of interests and needs (Example 10)’.

Hence, slack times profoundly shape the way technology scouts interact with their internal counterparts and how they manage knowledge across boundaries. While prior studies have shown that boundary spanners who effectively combine boundary spanning and managing constituents and tensions within organizations lead to superior outcomes (Carlile, 2004; Mudambi and Swift, 2009), we suggest that performance above aspirations may allow technology scouts to engage in these value-added activities (such as mitigating tensions between internal and external activities) in the first place. We propose:

Proposition 3d: In slack times, when spanning internal boundaries, technology scouts are more likely to emphasize the translation and transformation of external knowledge for the MNC.

CONCLUSION

Scholars have emphasized that spanning organizational and national boundaries to access external knowledge has increased considerably in recent decades (Bierly et al., 2009; Monteiro and Birkinshaw, 2017; Rothaermel and Alexandre, 2009). An important role in accessing external knowledge has been ascribed to boundary spanners, who mediate between the MNC's external environment and relevant actors (e.g., R&D units) inside the organization (Tushman and Katz, 1980). Many MNCs have formed dedicated units (e.g., technology scouts), not only to systematically span organizational and national boundaries to tap into external knowledge but also to span internal boundaries to make external knowledge available within the organization (Gassmann and Gaso, 2004; Monteiro, 2015) and researchers have also highlighted that such boundary-spanning processes are associated with a number of important performance benefits (Doz et al., 2001; Tushman and Katz, 1980). Surprisingly, however, much less scholarly attention has focused on the reverse relationship – that is, how a firm's prior performance affects boundary-spanning activities and, in particular, the activities of technology scouts.

To address this question, we have situated technology scouts in either harsh times (firm performance below aspirations) or slack times (performance above aspirations) and have presented several propositions related to how boundary-spanning activities are likely to unfold under those circumstances. In particular, we have focused on an emergent stream of research examining what boundary spanners in MNCs, particularly technology scouts, actually do (e.g., Gassmann and Gaso, 2004; Monteiro, 2015; Monteiro and Birkinshaw, 2017). Table I summarizes our predictions of how harsh and slack times ultimately affect whether, when and how technology scouts span organizational, national and internal boundaries.

Our theoretical model makes several contributions to the literature. First, our framework provides a behavioural explanation for the triggering mechanisms for different types of boundary-spanning processes in an MNC and suggests that both harsh and slack times may give rise to the inception of a boundary-spanning unit in the first place. The study highlights that while the link between boundary spanning and performance has been established by prior literature, how performance ultimately shapes boundary-spanning activities in global organizations is an equally important area of inquiry.

We propose that, in both harsh and slack times, technology scouts increase the intensity of crossing organizational boundaries, but for very different reasons. While in harsh times, they are motivated by a problem to seek external knowledge, in slack times, they may opportunistically consider 'next problems'. This distinction has implications for a third category of MNCs: those located around the aspiration point with neither the pressure nor the short-term slack resources to cross organizational boundaries. Figure 1 provides a graphical representation of how the intensity of organizational boundary spanning by technology scouts may be shaped by performance aspiration comparisons.

Second, we suggest that harsh and slack times impose important conditions on how technology scouts balance their boundary-spanning activities, in particular, when they span national boundaries to tap into foreign knowledge. We expect that in harsh times, problem-oriented motives may encourage spanning national boundaries, but limited

Table I. Summary of propositions of how harsh and slack times shape technology scouts’ boundary-spanning behaviour

<i>Boundary Spanned</i>	<i>Activity</i>	<i>Harsh times</i>	<i>Slack times</i>
Organiza-tional	Technology scouts span organizational boundaries to tap into external sources of knowledge.	Technology scouts intensify boundary spanning the more performance falls below MNC’s aspiration. (P1a) (Example 1) Increase in likelihood of boundary-spanning unit being formalized and incepted (P1b) (Example 2)	Technology scouts intensify boundary spanning intensity the more performance rises above MNC’s aspiration. (P1c) (Example 3a, b) Increase in likelihood of boundary-spanning unit being formalized and incepted (P1d) (Example 4)
National	Technology scouts span national boundaries to reach ‘best in the world’ knowledge in foreign locations.	Close to Aspiration Point: Lower boundary-spanning intensity by technology scouts (P1e, P1f) Technology scouts engage in punctuated national boundary spanning (P2a) (Example 5,6) Technology scouts have a predefined and narrow scope to span national boundaries. (P2b) (Example 6)	Technology scouts more likely span national boundaries (in addition to activities at home). (P2c) (Example 7) Technology scouts have broad and open mandates to span national boundaries (P2d) (Example 7)
Internal	Technology scouts span internal boundaries to R&D and business units to present and mobilize external opportunities within the organization.	Technology scouts more likely present external knowledge as a substitute for internal R&D activities. (P3a) (Example 8) Technology scouts emphasize the transfer and channeling of knowledge into the MNCs. (P3b) (Example 8)	Technology scouts more likely present external knowledge as a complement for internal R&D activities. (P3c) (Example 9) Technology scouts emphasize the translation and transforming of knowledge for use within the MNCs. (P3d) (Example 10)

resources in harsh times lead to trade-offs for technology scouts to span national boundaries and maintain their activities inside their home location. As a result, spanning national boundaries in harsh times will be more likely one-off and more directed. Conversely, in slack times, technology scouts have a more open mandate to engage in national boundary spanning without having to make compensatory trade-offs. This perspective adds fresh insights to the ambidexterity debate by considering performance as an important input (not output) of spanning boundaries in an ambidextrous way (Raisch et al., 2009).

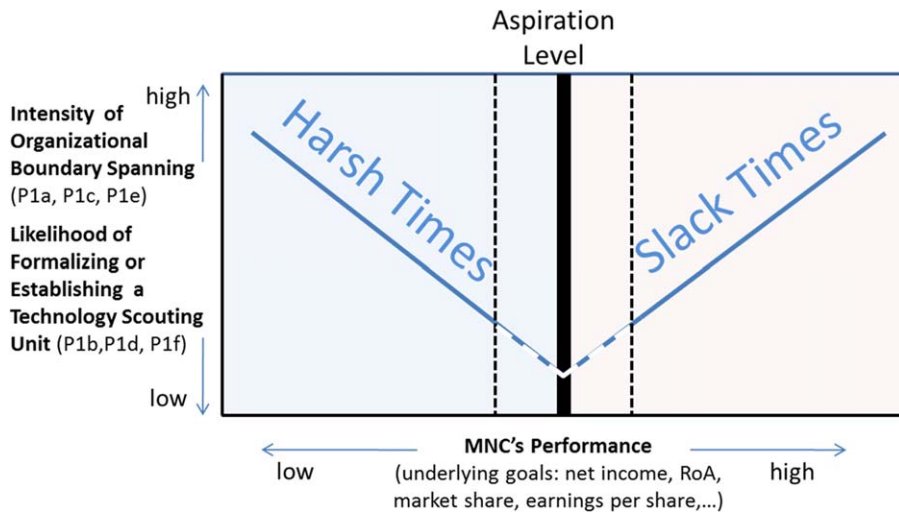


Figure 1. Global boundary spanning by technology scouts induced by harsh and slack times. [Colour figure can be viewed at wileyonlinelibrary.com]

Third, we have revealed important behavioural clues as to how technology scouts span internal boundaries to connect and engage with their internal constituents in harsh and slack times. We identified that, when spanning internal boundaries in harsh times, technology scouts are more likely to present external solutions as substitutes to internal R&D, while in slack times, technology scouts are more likely to showcase external solutions as complements to internal R&D. In a related vein, we identified that, due to their limited resources, technology scouts in harsh times are likely to emphasize the channeling and transferring of external knowledge into the organization, while we expect them in slack times to focus on the translation and transformation of external knowledge for their internal constituents (Monteiro and Birkinshaw, 2017). This extends prior work that suggested that value-adding, boundary-spanning activities such as transformation allow for establishing common interests and managing tensions across boundaries, which can ultimately improve an organization's competitive advantage, in particular when knowledge at the boundaries is characterized by high novelty (Carlile, 2004; Mudambi and Swift, 2009; Tippman et al., 2017). We reveal that such value-adding behaviour by technology scouts may, in fact, be affected by performance feedback and is more likely to emerge in slack times.

Finally, the study expands prior work related to the behavioural theory of the firm. While researchers have long considered harsh and slack times as important contingencies for broad activities such as R&D spending (Chen and Miller, 2007; Greve, 2003a), our paper focuses on a particular type of organizational activity (boundary spanning) and provides an in-depth look at the behaviour of specific actors (technology scouts) when spanning organizational, national and internal boundaries. This focus allows us to move away from the organizational level typically used in behavioural studies and get closer to how harsh and slack times affect the actual behaviour of individuals within MNCs. Hence, the paper connects the behavioural theory of the firm to the growing

literature on boundary spanners in MNCs (such as technology scouts or corporate executives) and the activities they perform (e.g., Birkinshaw et al., 2017; Monteiro and Birkinshaw, 2017).

Our theoretical model, applying performance feedback as an input to boundary-spanning activities, could be extended in several ways. First, throughout the paper, we have made the assumption that technology scouts deal with knowledge with some elements of novelty, i.e., knowledge that represents 'something new and different' for the organization or 'departs significantly from existing product architecture' (Carlile and Reberntsch, 2003, p. 1184). However, recent research (Monteiro, 2015) has shown that sometimes an MNC goes around the globe only to end up paying attention to technologies that are very close to its existing knowledge base. In this paper, we have not differentiated opportunities with respect to their extent of novelty and how this might affect boundary-spanning behaviour by technology scouts in harsh and slack times. Rather, our focus is on how different environmental conditions (i.e., being in harsh or slack times) determine different boundary-spanning behaviours. Our findings open new research avenues, for instance, by combining the propositions we developed here with Carlile's (2004) framework for managing of knowledge across boundaries in the context of product innovations, which distinguishes three key boundaries with increasing complexity (syntactic, semantic and pragmatic) and three key processes by boundary spanners (transfer, translation and transformation). More deeply examining the type of external opportunities, possibly at different levels of novelty, and further connecting this literature with that on the behavior and processes performed by technology scouts (Monteiro and Birkinshaw, 2017) may be a promising path for further research.

Second, we have separately discussed three distinct activities of technology scouts related to organizational, national, and internal boundaries. However, technology scouts often engage in these activities simultaneously, and opportunities and challenges from the process stem from technology scouts embeddedness in both an external (e.g., their activities with firms outside the organization) and internal contexts (e.g., the activities by which they engage with actors inside the organization) (Meyer et al., 2011; Monteiro and Birkinshaw, 2017). While our paper makes several suggestions as to how harsh and slack times affect technology scouts' attempts to balance boundary-spanning activities (e.g., compensatory trade-offs stemming from spanning national boundaries in harsh times), future research could more systematically examine how technology scouts concomitantly engage in all the outlined activities in harsh and slack times. To that end, it may become necessary to once more consider that boundary spanning is not necessarily a differentiated and defined function but may require several actors in different boundary-spanning roles (Friedman and Podolny, 1992). For instance, some technology scouts may focus on the generation of external opportunities, while others focus on mobilizing such opportunities within the organization. Examining these processes in more detail may also allow a more systematic investigation of how, for example, the intensity of internal boundary spanning unfolds or how technology scouts engage with external constituents (potential knowledge sources) in harsh and slack times, which we did not uncover in the paper. We believe the way in which a division of labour in boundary spanning is effectively set up in global organizations and how the setup is shaped by harsh and slack times are questions for future research.

Finally, we have emphasized global boundary spanning as a way to tap into external and foreign knowledge. However, gaining access to such knowledge often requires reciprocity, i.e., revealing knowledge to potential external partners (Monteiro et al., 2008; Monteiro and Birkinshaw, 2017). While we have characterized technology scouts in general as knowledge-seeking, one could easily imagine them also transmitting knowledge to their external constituents (e.g., to help them prepare for submitting knowledge to the MNC, or sharing information in general). We would conjecture that, over time, technology scouts trying to access external knowledge without revealing their MNC's knowledge would compromise trust between parties, and therefore, the MNC's boundary-spanning efforts will become increasingly difficult (Monteiro et al., in press). It would be interesting to expand our study's activities by technology scouts to include how they interact with their external environment (e.g., emphasizing secrecy vs. being open when engaging with external partners) and how this behaviour may be shaped by harsh and slack times.

Overall, we believe that global boundary spanning has become an increasingly important activity for MNCs in many industries. Our study illustrates the behaviour of boundary spanners engaged in this process in the form of technology scouts. We reveal that harsh and slack times may be important triggers for whether, when and how technology scouts ultimately span boundaries. We hope this paper will inspire more research relating to the micro-level behaviour of decision makers and technology scouts in MNCs and how their activities may be triggered by macro-organizational factors such as harsh and slack times. We believe that behavioural arguments, in particular, theories of performance feedback, may offer fertile ground for future research to make such connections.

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NOTES

- [1] We make the important assumption that technology scouts deal with knowledge with some elements of novelty, i.e., knowledge that represents 'something new and different' for the organization or 'departs significantly from existing product architecture' (Carlile and Rebentisch, 2003, p. 1184). Carlile's (2004) framework for managing knowledge across boundaries when innovation is desired suggests that novelty leads to increasing complexity of knowledge boundaries (e.g., due to the specialization of actors in technological areas or their mutual dependence on each other to ultimately make use of the knowledge). Such highly complex knowledge boundaries are exactly the type of boundaries we expect technology scouts to span. In our conclusion, we suggest future areas of research that could explicitly differentiate degrees of novelty of external opportunities and boundary spanning by technology scouts.
- [2] For the purpose of clarity, our paper separately examines these three distinct activities by technology scouts. However, we acknowledge that technology scouts engage in all of these activities concomitantly, and that opportunities and challenges from the process stem from technology scouts being simultaneously embedded in external, foreign and corporate contexts (Meyer et al., 2011; Monteiro and Birkinshaw, 2017). We address this in the discussion section.

- [3] Fleming and Waguespack (2007) noted that ‘although the measures of boundary spanning and brokerage correlate empirically, the concepts remain theoretically distinct’.
- [4] We acknowledge that this is an extreme representation of boundary spanners. Indeed, researchers have explained that boundary spanning is not necessarily a differentiated function and boundary-spanning roles may be performed by different actors (Friedman and Podolny, 1992). In a similar vein, technology scouts may perform differentiated tasks, including the generation of opportunities and mobilizing such opportunities within the organization. We expand on these ideas in the discussion section.
- [5] For the purpose of our paper, managers can be on the level of the top management team or at the business unit level. Importantly, these managers directly assign mandates to boundary-spanning units such as technology scouts.
- [6] Our arguments are based on financial performance dimensions (e.g., return on assets or net income), which follows Cyert’s and March’s original idea to consider metrics that directly affect decision-makers. However, Cyert and March also considered non-financial performance dimensions, including production and inventory, and market share goals. It is also likely that managers take into consideration innovative performance – that is, its innovative output and how it compares to industry peers. While we believe our theoretical framework applies to non-financial performance measures, we leave such an exploration for future research.
- [7] We acknowledge that there are other non-behavioural explanations which are likely to affect an MNC’s global boundary-spanning activities (e.g., prior experience). While we are cognizant of the importance of those alternative explanations, our goal is to reveal the unique importance of performance feedback to global boundary-spanning activities and technology scouts specifically.
- [8] Each interview lasted on average 45 minutes. In prior work, we also documented a detailed description of the technology-scouting process and some of its fundamental tensions in a multinational organization (Klueter and Monteiro, 2014).
- [9] BTF research has long indicated that absolute performance and adverse events (e.g., project withdrawals, railway accidents, or space-launch accidents) may substantially alter an organization’s behaviour (Madsen and Desai, 2010). We expect very high and very low absolute performance to be incorporated into the aspirations of organizational decision makers so that our theoretical framework also largely applies in such events.
- [10] We are excluding extreme cases of poor performance. Firms performing very poorly (i.e., verging on bankruptcy) behave differently, as managers shift their attention to survival, which leads these firms to search less (Chen and Miller, 2007).
- [11] While, from this perspective, slack times enable firms to engage in and intensify global boundary-spanning activities, we cannot make claims that such behaviour ultimately leads to innovation. Research has emphasized that the direct relationship of slack to innovation outcomes remains subject to debate (Nohria and Gulati, 1996; Pitelis, 2007).
- [12] The aforementioned studies revealed that the overwhelming proportion of R&D activity is within the MNC’s home countries (up to 90 per cent). We hence formulate our propositions based on crossing national boundaries, i.e., tapping into knowledge beyond the firm’s home country. However, we acknowledge that MNCs have subsidiaries located around the globe and we believe our theorizing could be extended and applied to spanning national boundaries to locations in which the MNCs are less active.
- [13] Once we understand that MNCs make choices to deliberately span national boundaries to tap into foreign knowledge, similar arguments can apply when considering different forms of knowledge proximity (e.g., geographical or technological). We expand on such alternatives and the necessity of incorporating alternative boundaries into our arguments in the discussion section. For the purposes of this paper, however, we limit ourselves to spanning national boundaries.

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